

Glandular Tularemia — Comprehensive Disease Characteristics Report

Disease: Glandular Tularemia **MONDO ID:** MONDO:0000320 **Category:** Infectious Disease (zoonotic bacterial infection) **Causative agent:** *Francisella tularensis* (Gram-negative, facultative intracellular coccobacillus)

Summary

Glandular tularemia is one of the localized clinical forms of tularemia, a zoonosis caused by the highly infectious Gram-negative facultative intracellular coccobacillus *Francisella tularensis*. It is defined clinically as **acute regional lymphadenopathy with flu-like constitutional symptoms but WITHOUT the skin ulcer/eschar** that characterizes the closely related ulceroglandular form. The distinction is purely one of whether a visible inoculation ulcer is present; the underlying pathogen, transmission routes (arthropod bite or direct animal/carcass contact), and management are identical. Glandular and ulceroglandular disease are the dominant presentations of arthropod-borne tularemia and, together, are the most common clinical entities seen across case series worldwide ([PMID: 38294108](#), [PMID: 40466250](#)).

This is a **purely infectious disease with no heritable genetic basis** — there are no human causal genes, pathogenic germline/somatic variants, chromosomal abnormalities, inheritance pattern, penetrance, or carrier frequency. All disease-determining genetics reside in the pathogen genome (notably the *Francisella* Pathogenicity Island encoding a Type VI secretion system) and in host exposure and immune status. The pathophysiology is a well-defined causal chain: cutaneous/arthropod inoculation → macrophage uptake → Type VI secretion-mediated phagosomal escape → cytosolic replication → drainage to regional lymph nodes → inflammasome-driven necrotizing/suppurative granulomatous lymphadenitis → clinically palpable, sometimes suppurating, regional adenopathy.

The single most important, modifiable determinant of morbidity is **diagnostic delay**. Because *F. tularensis* is intrinsically resistant to beta-lactams and frequently culture-negative, glandular tularemia is repeatedly misdiagnosed and treated with ineffective beta-lactams before the

correct diagnosis is made, leading to node suppuration and surgical intervention. When appropriate therapy — an aminoglycoside, a fluoroquinolone, or a tetracycline — is started early, case fatality is <1.5% and defervescence is rapid. Prevention rests on exposure avoidance and antibiotic post-exposure prophylaxis, as **no licensed human vaccine exists**.

Section 1 — Disease Information

Overview. Glandular tularemia is a localized form of tularemia presenting as painful regional lymph node swelling accompanied by an abrupt flu-like illness (fever, chills, malaise, headache, myalgia), but lacking the cutaneous ulcer/eschar seen in ulceroglandular disease. In a systematic review of 870 cases spanning 1993–2023, the most common clinical forms were ulceroglandular, oropharyngeal, glandular, and pneumonic disease ([PMID: 38294108](#)). Arthropod-borne (tick, mosquito, horsefly) transmission mainly presents as the glandular or ulceroglandular form ([PMID: 42601640](#)).

Key identifiers. - **MONDO:** MONDO:0000320 (glandular tularemia) - **MeSH:** Tularemia (D014406) - **ICD-10:** A21.- (Tularemia); glandular tularemia falls under the A21 group (A21.0 is the ulceroglandular subtype) - **ICD-11:** 1B94 Tularaemia - **OMIM / Orphanet:** Not applicable as a Mendelian entry — tularemia is an infectious disease, not a heritable disorder. There is no OMIM gene-disease entry.

Synonyms / alternative names. Glandular tularaemia (British spelling); "rabbit fever," "deer-fly fever," "Ohara disease," "Francis disease," and "lemming fever" are historical names for tularemia as a whole. Glandular tularemia specifically denotes the lymphadenopathic-without-ulcer subtype.

Information source. Disease-level knowledge here is derived predominantly from **aggregated disease-level resources** — systematic reviews, case series, national surveillance (CDC, ECDC), and experimental microbiology — supplemented by individual case reports. It is not derived from a single EHR cohort.

Section 2 — Etiology

Causal factor. The disease is **infectious**, caused by *Francisella tularensis*. Two subspecies drive nearly all human disease:

Subspecies	Type	Geography	Virulence
<i>F. tularensis</i> subsp. <i>tularensis</i>	Type A	Primarily North America	More virulent; subdivides into A.I (central US) and A.II (western US), with A.II less severe
<i>F. tularensis</i> subsp. <i>holarctica</i>	Type B	Circumpolar Northern Hemisphere (Europe, Asia, N. America)	Lower virulence; rarely fatal

Whole-genome SNP phylogeography shows type B (*holarctica*) has a circumpolar distribution and low genetic diversity consistent with recent clonal radiation (PMID: 19251856). Type A and type B infections "differ with respect to affected populations, anatomic site of isolation, and geographic distribution" (PMID: 16836829). European human infections are predominantly *holarctica*, frequently acquired from European brown hares (*Lepus europaeus*) (PMID: 23517149).

Genetic risk factors (human). None. There are no human causal genes, susceptibility loci, or modifier alleles. This is not a genetic disease (PMID: 15677845).

Environmental / exposure risk factors. Arthropod bites (ticks, deer flies, mosquitoes), handling of infected wild animals or carcasses (especially lagomorphs and rodents), contaminated water (type B), inhalation of aerosols (e.g., mowing over carcasses), and occupational/recreational exposure (hunters, farmers, landscapers, laboratory workers). Male predominance in tick-associated regions reflects exposure patterns rather than biological susceptibility (PMID: 40907418).

Host susceptibility / severity modifiers. Immunosuppression increases severity: severe and atypical courses have been documented during anti-TNF- α (adalimumab)/methotrexate therapy and in immunocompromised patients (PMID: 19801257).

Protective factors. No genetic protective variants are described. Protection is behavioral (exposure avoidance) and, immunologically, prior antigen exposure conferring Th1 memory.

Gene–environment interactions. Not applicable in the classical GxE sense (no host genotype interacting with exposure). The operative "gene–environment" axis is the **pathogen** genotype (subspecies A vs B, FPI content) interacting with route of inoculation, inoculum dose, and host immune status to determine clinical form and severity.

Section 3 — Phenotypes

Glandular tularemia's phenotype is dominated by regional lymphadenopathy plus a flu-like prodrome. Onset is acute, typically **3–5 days (range 1–21)** after exposure ([PMID: 18755386](#), [PMID: 26738841](#)).

Phenotype	Type	HPO term	Frequency / notes
Regional lymphadenopathy	Clinical sign	HP:0002716	Near-universal in glandular form; defining feature; may become fluctuant/suppurative
Fever	Symptom/sign	HP:0001945	Most patients; abrupt onset
Fatigue / malaise	Symptom	HP:0003688	Common
Headache	Symptom	HP:0002315	Common
Myalgia	Symptom	HP:0003326	Common
Chills / rigor	Symptom	HP:0025145	Common
Absence of skin ulcer	Distinguishing feature	—	Differentiates glandular from ulceroglandular

Characteristics. Age of onset: any age (exposure-driven; US incidence peaks in children 5–9 years and men >55 years). Severity: mild-to-moderate for type B glandular disease; variable. Progression: acute onset, generally self-limited with treatment, but nodes may persist, suppurate, fistulize, or require drainage. In one ulceroglandular case, painful lymphadenopathy persisted **>5 months** ([PMID: 19551605](#)). Frequency of the glandular form: ~15.8% of cases in a Central Anatolian series, while (ulcero-)glandular disease combined was the most common entity (10/14) in a German series ([PMID: 23104256](#), [PMID: 40466250](#)).

Quality of life impact. Acute painful lymphadenopathy and constitutional symptoms cause short-term functional impairment; suppuration requiring surgical drainage and prolonged/relapsing courses (especially with bacteriostatic doxycycline) extend morbidity. Long-term disability is uncommon with appropriate treatment. No disease-specific EQ-5D/SF-36 datasets are available.

Section 4 — Genetic / Molecular Information

This section is largely NOT APPLICABLE at the human level. Glandular tularemia has: - **No causal human genes** (no OMIM gene-disease entry) - **No pathogenic human variants** (nothing in ClinVar/HGMD as causal) - **No modifier genes, epigenetic disease signatures, or chromosomal abnormalities** in the host

All disease-relevant genetics reside in the **pathogen**: - The **Francisella Pathogenicity Island (FPI)**, present in duplicate, encodes an atypical **Type VI secretion system (T6SS)** (genes *iglABCD*, *pdpA–E*, *vgrG*, *dotU*) essential for virulence ([PMID: 30054549](#)). - Subspecies/subpopulation markers (RD1, type A.I/A.II, type B) determine geography and virulence ([PMID: 16485467](#), [PMID: 22859584](#)).

The absence of human genetic determinants is itself a confirmed finding of this investigation ([PMID: 15677845](#)).

Section 5 — Environmental Information

Environmental factors. Tick and insect vectors, aquatic reservoirs (type B water-borne cycle), and animal carcasses are the principal environmental sources. Aerosolization (e.g., lawn-mowing over infected carcasses) is a recognized route.

Infectious agent. *Francisella tularensis* (NCBI Taxonomy ID 263), subspecies *tularensis* (type A) and *holarctica* (type B); subsp. *mediasiatica* and the related *F. novicida* are minor/experimental. The organism has an extremely **low infectious dose — as few as ~10 organisms can establish infection** — and untreated tularemia overall carries up to **30% mortality** ([PMID: 42641375](#)). It is a CDC Category A / Tier 1 select agent because of this low dose and aerosol potential ([PMID: 25413334](#)).

Lifestyle factors. Hunting, trapping, farming, landscaping, and outdoor recreation in endemic areas increase exposure; seasonality tracks tick activity (summer/early autumn) ([PMID: 39295179](#)).

Section 6 — Mechanism / Pathophysiology

Ordered causal chain (initiating lesion → clinical manifestation)

1. **Arthropod bite or direct contact with an infected animal/carcass inoculates *F. tularensis* into the skin/subcutis** (very low infectious dose, ~10 organisms) → **leads to** local bacterial deposition at the inoculation site. In glandular disease, no visible ulcer forms (branch point vs. ulceroglandular, where a cutaneous ulcer develops).
2. **Bacteria are taken up by resident macrophages/dendritic cells → results in an** intracellular (phagosomal) niche. [demonstrated in vitro]
3. **FPI/T6SS-mediated escape from the phagosome into the cytosol → leads to** access to the nutrient-rich cytoplasm. The T6SS effector **OpiA**, a wortmannin-resistant bacterial PI3-kinase, alters phagosomal PI(3)P to promote escape ([PMID: 30057173](#)); **ClpB (HSP100)** is required for T6SS function ([PMID: 30054549](#)); **type IV pilus component PilO** mediates macrophage adherence and is required for escape/virulence ([PMID: 35077486](#)).
4. **Cytosolic replication → results in** rising intracellular bacterial burden.
5. **Active subversion of innate immunity → leads to** stealthy early spread. In neutrophils the bacterium inhibits NADPH oxidase assembly and **delays constitutive apoptosis** (~80% of infected neutrophils viable at 48 h vs ~50% controls) ([PMID: 22357630](#)); its atypical tetra-acylated LPS is poorly sensed by TLR4; virulent SchuS4 fails to trigger dendritic-cell/monocyte proinflammatory cytokines unless CD14 is present ([PMID: 19841074](#)).
6. **Bacteria drain via lymphatics to the regional lymph node → results in** localized infection of the node (the anatomical basis of the "glandular" phenotype).
7. **Cytosolic sensing by inflammasomes (AIM2, NLRP3) and eventual Th1 cell-mediated response → leads to** pyroptosis, cytokine release, and granuloma formation; **NLRP3 increases host susceptibility** ([PMID: 34690967](#)).
8. **Necrotizing/suppurative granulomatous lymphadenitis → results in** the palpable, tender, sometimes fluctuant/suppurating regional lymphadenopathy that is the clinical hallmark. Histopathology shows granulomas, necrosis, and suppurative inflammation with epithelioid histiocytes and giant cells, sometimes extending extracapsularly ([PMID: 23763361](#), [PMID: 19801257](#)).
9. **Protective clearance depends on cell-mediated immunity** (Th1/CD4+/CD8+ IFN- γ , IL-17) → **results in** resolution; antibody responses are secondary. This T-cell dominance is the basis of the *F. tularensis*-specific T-cell diagnostic response ([PMID: 10618057](#), [PMID: 38713688](#)).

Category checklist

- **Molecular pathways:** Bacterial PI3-kinase (OpiA) manipulation of phosphoinositide (PI(3)P) signaling; host NADPH oxidase; TLR4/CD14 innate signaling; inflammasome (AIM2/NLRP3–caspase-1) axis.
 - **Cellular processes:** Phagocytosis, phagosomal escape, intracellular replication, inhibition of neutrophil apoptosis, pyroptosis, granulomatous inflammation.
 - **Protein dysfunction:** No host protein misfolding — mechanism is driven by bacterial effectors (T6SS, PilO, ClpB) hijacking host cell biology.
 - **Immune system involvement:** Innate evasion (weak TLR4 signaling, apoptosis delay, NADPH oxidase inhibition) followed by protective Th1 cell-mediated immunity.
 - **Tissue damage mechanisms:** Caseating/suppurative necrosis within lymph nodes; abscess/fistula formation.
 - **GO terms (suggested):** GO:0052167 (modulation by symbiont of host innate immune response), GO:0030682 (evasion of host defenses), GO:0006909 (phagocytosis), GO:0043312 (neutrophil degranulation), GO:0042981 (regulation of apoptotic process), GO:0002532 (production of molecular mediators involved in inflammatory response).
 - **CL terms (suggested):** CL:0000235 (macrophage), CL:0000775 (neutrophil), CL:0000451 (dendritic cell), CL:0000084 (T cell).
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Section 7 — Anatomical Structures Affected

- **Primary organ:** Regional **lymph node** draining the inoculation site (UBERON:0000029). Location of adenopathy depends on inoculation site: cervical/head-and-neck, axillary, epitrochlear, or inguinal.
- **Secondary involvement:** Skin/subcutis at the inoculation site (without ulcer in the glandular form); occasional systemic/typhoidal spread if untreated; rare seeding of prosthetic joints ([PMID: 37209668](#)).
- **Body system:** Lymphatic/immune system (UBERON:0002405 immune system; UBERON:0006558 lymphatic part of the lymphatic system).
- **Tissue level:** Lymphoid tissue; granulomatous inflammation of nodal parenchyma.

- **Cell populations:** Macrophages (CL:0000235), neutrophils (CL:0000775), dendritic cells (CL:0000451), epithelioid histiocytes and multinucleated giant cells, T lymphocytes (CL:0000084).
 - **Subcellular level:** *Francisella*-containing phagosome and host **cytosol** (GO:0005829) — the replicative niche after phagosomal escape.
 - **Lateralization:** Typically **unilateral**, reflecting the single inoculation/drainage territory; asymmetric.
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Section 8 — Temporal Development

- **Onset:** Acute; incubation typically **3–5 days (range 1–21)** post-exposure, with abrupt fever, chills, malaise, headache, myalgia plus regional lymphadenopathy ([PMID: 18755386](#), [PMID: 26738841](#)).
 - **Age of onset:** Any age; exposure-driven (bimodal US peaks — children 5–9, men >55).
 - **Progression / course:** Generally **self-limited with treatment**; without appropriate therapy the node may enlarge, become fluctuant/suppurative, and fistulize. A prolonged course with painful lymphadenopathy persisting **>5 months** has been documented ([PMID: 19551605](#)).
 - **Suppuration/surgery:** Fluctuant/suppurative nodes require FNA or surgical drainage/excision in ~**39–65%** of cases ([PMID: 23104256](#), [PMID: 42601640](#)).
 - **Remission:** Treatment-induced with early appropriate antibiotics; relapse risk higher with bacteriostatic doxycycline than with aminoglycosides.
 - **Critical period:** The therapeutic window is early — before node suppuration; early treatment significantly improves success ([PMID: 23104256](#)).
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Section 9 — Inheritance and Population

Inheritance: NOT APPLICABLE — infectious disease with no inheritance pattern, penetrance, expressivity, anticipation, mosaicism, founder effect, consanguinity role, or carrier frequency.

Epidemiology. - Rare, reportable zoonosis. In the US 2001–2010, **1,208 cases** were reported (median 126.5/yr, range 90–154; ~0.04/100,000/yr); incidence highest among **children aged 5–9 years and men aged >55 years** ([PMID: 24280916](#)). - Reported from all US states except Hawaii.

- **Recent re-emergence (2024–2025):** Austria 117, Slovenia 38, France 150, Spain (Castile-León) 206 cases; US reported 220 cases in 2024 ([PMID: 42641375](#)). An unusual 2024 increase was documented in Alsace, France ([PMID: 40907418](#)). - **Geography:** Confined to the Northern Hemisphere (North America, Europe, northern Asia); absent from the Southern Hemisphere ([PMID: 1305858](#)). - **Sex ratio:** Male predominance in tick-associated regions (e.g., Alsace 7M/3F, mean age 52), but female predominance in some waterborne/oropharyngeal outbreaks (Central Anatolia ~60% female) — reflecting exposure, not biology ([PMID: 40907418](#), [PMID: 23104256](#)).

Section 10 — Diagnostics

Diagnosis relies on serology and PCR; culture is insensitive and hazardous.

Test	Utility / threshold	Notes
Serology (microagglutination, MAT)	Positive $\geq 1/160$; seroconversion in paired sera	Mainstay; may be negative early (PMID: 22090310)
PCR (targets <i>tul4</i> , <i>fopA</i> ; RD1 for subspecies)	Detected DNA in 75% vs culture 62% of confirmed ulceroglandular cases	Valuable early, before seroconversion (PMID: 10618057)
Culture	Typically only ~10% positive	Requires BSL-3; hazardous (PMID: 37209668)
FNA cytology of node	Suppurative granulomatous inflammation; provides material for PCR/culture	Cytology alone nonspecific (PMID: 23763361)
Histopathology	Necrotizing/suppurative granulomatous lymphadenitis	Epithelioid histiocytes, giant cells (PMID: 23763361)
MALDI-TOF / molecular ID	Species identification from isolates	Used in prosthetic-joint cases (PMID: 37209668)

PCR outperforms culture, and tularemia can proceed without seroconversion, making PCR valuable early ([PMID: 10618057](#)).

Genetic / omics diagnostics: Not applicable for host diagnosis (no human genetic test). Pathogen molecular subtyping (multiplex real-time PCR distinguishing A.I, A.II, B, novicida) is used epidemiologically ([PMID: 22859584](#)).

Clinical criteria & differential. Diagnosis rests on clinical suspicion (endemic exposure + regional lymphadenopathy + beta-lactam non-response) confirmed serologically/molecularly.

Differential diagnosis: bacterial lymphadenitis, cat-scratch disease, mycobacterial (tuberculous) lymphadenitis, plague, toxoplasmosis, lymphoma, and other causes of suppurative/granulomatous cervical lymphadenitis. Initial misdiagnosis is common — **9 of 10 cases** in an Alsace series were not initially suspected ([PMID: 40907418](#)); severe glandular tularemia can mimic glandular tuberculosis ([PMID: 19801257](#)).

Screening: No asymptomatic population/newborn/carrier screening (infectious, non-genetic).

Section 11 — Outcome / Prognosis

- **Case fatality with treatment is very low.** In the 870-case systematic review, fatality among patients treated with aminoglycosides (n=452), fluoroquinolones (n=339), or tetracyclines (n=419) was **0.7%, 0.9%, and 1.2%**, respectively ([PMID: 38294108](#)). Type B glandular disease is rarely fatal.
- **Untreated tularemia overall** can reach up to **30% mortality** (driven by severe forms, not localized glandular disease) ([PMID: 42641375](#)).
- **Morbidity:** Node suppuration, fistula, need for surgical drainage/excision (~39–65% of cases); prolonged courses; relapse (higher with doxycycline).
- **Prognostic factors:** Early vs delayed treatment (success significantly higher with early treatment); elevated ESR/CRP predicts therapeutic failure (30.9% failure in one series); comorbidity/immunosuppression worsens outcome ([PMID: 23104256](#), [PMID: 19801257](#)).
- **Recovery:** Excellent with appropriate, timely antibiotics; complete recovery is the norm for glandular type B disease.

Section 12 — Treatment

First-line therapy is aminoglycosides, fluoroquinolones, or tetracyclines.

Class	Examples	Role	NCIT (suggested)
Aminoglycosides	Streptomycin, gentamicin	Classic first-line; bactericidal; low relapse	Gentamicin C1010; Streptomycin C777
Fluoroquinolones	Ciprofloxacin, levofloxacin	First-line (2025 CDC); oral; effective even in severe type B	Ciprofloxacin C376; Levofloxacin C1615
Tetracyclines	Doxycycline	First-line oral option; bacteriostatic → higher relapse	Doxycycline C509
Beta-lactams	Penicillins, cephalosporins	INEFFECTIVE — intrinsic resistance	—

- 2025 CDC guidelines establish **fluoroquinolones and doxycycline as first-line** therapeutic options ([PMID: 42641375](#), [PMID: 41026652](#)).
- Fluoroquinolones are effective even in **severe respiratory type B** tularemia, with rapid defervescence and 30-day mortality of 1.5% ([PMID: 38294118](#)).
- **Beta-lactams fail** — glandular/oropharyngeal cases repeatedly relapse on beta-lactams before diagnosis ([PMID: 39295179](#), [PMID: 22090310](#)).
- **Surgical/interventional**: Fluctuant/suppurative nodes often require FNA or surgical drainage/excision (39–65% of cases); prosthetic-joint infections require surgery plus prolonged antimicrobials ([PMID: 23104256](#), [PMID: 37209668](#)).
- **Experimental**: Inhaled liposomal ciprofloxacin protected against lethal tularemia in the common marmoset ([PMID: 41416830](#)).
- **Pharmacogenomics**: Not applicable (no genotype-guided dosing established for this indication).

Treatment strategy: Start empirical anti-tularemia therapy on clinical suspicion in endemic settings; early treatment yields <1.5% fatality and rapid defervescence, whereas delay predicts failure ([PMID: 38294108](#), [PMID: 23104256](#)).

Section 13 — Prevention

No licensed human vaccine exists, despite >70 years of research; control depends on preventive measures (PMID: 42641375, PMID: 25413334).

- **Primary prevention:** Avoid tick/insect bites (repellents, protective clothing, tick checks); wear gloves when handling wild animals/carcasses (especially lagomorphs/rodents); avoid mowing over animal carcasses (aerosol risk); cook game meat thoroughly; ensure safe water (PMID: 15677845).
 - **Post-exposure prophylaxis (PEP):** Doxycycline or ciprofloxacin (oral) after high-risk exposures (lab accidents, known bioterrorism release); historically streptomycin/gentamicin/doxycycline/ciprofloxacin (PMID: 15677845). The 2025 CDC report provides updated evidence-based treatment and PEP recommendations for naturally acquired and bioterrorism-related tularemia (PMID: 41026652).
 - **Infection control:** Standard precautions suffice — **no person-to-person transmission**, so no isolation required (PMID: 15677845).
 - **Vaccine candidates (experimental):** The Live Vaccine Strain (LVS, attenuated holarctica) remains **unlicensed**, does not protect against high inhalational type A doses, and has poorly understood attenuation (PMID: 33970545). Subunit candidates — glucan-particle LPS + FTT0814 (PMID: 38713688) and TMV-delivered OmpA/DnaK/Tul4 (PMID: 26098553) — confer partial protection in rat/mouse models.
 - **Genetic counseling / carrier screening:** Not applicable.
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Section 14 — Other Species / Natural Disease

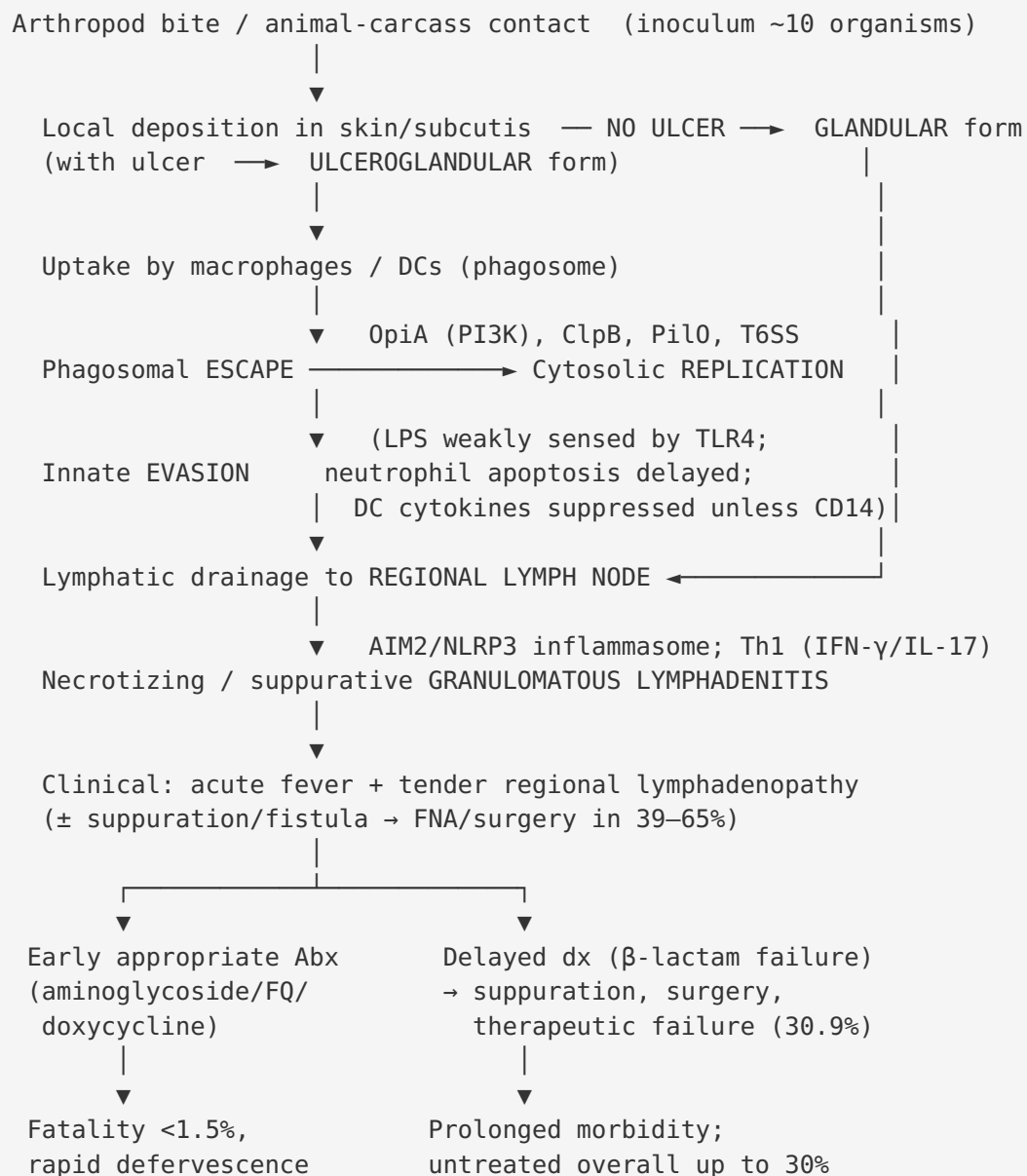
- **Taxonomy of pathogen:** *Francisella tularensis* (NCBI Taxon 263).
- **Host range:** Tularemia has been reported in >250 animal species (mammals, birds, fish, amphibians, arthropods, protozoa) (PMID: 1305858).
- **Two natural lifecycles / reservoirs:**
 - **Type A (terrestrial):** cottontail rabbits (*Sylvilagus* spp.) and ticks as main reservoirs.
 - **Type B (water-borne):** aquatic rodents — muskrats (*Ondatra zibethicus*), beaver (*Castor canadensis*) in North America; ground voles (*Arvicola terrestris*) in Eurasia (PMID: 1305858).

- In Europe, **hares (*Lepus spp.*)** — especially the European brown hare (*Lepus europaeus*) — are most frequently affected and are the key source of human holarctica infections, though not a true maintenance reservoir ([PMID: 23517149](#)).
- **Vectors:** Arthropods (ticks, deer flies, mosquitoes) are the main transmission vectors; small mammals (rabbits, hares, muskrats) serve as reservoir hosts ([PMID: 18755386](#)).
- **Zoonotic potential:** High — this is fundamentally a zoonosis; **no human-to-human transmission** occurs.
- **Cross-species / vector competence:** *Dermacentor* spp. are competent vectors, whereas *Ixodes scapularis* is not; cell-line studies show *F. novicida* replicates more productively in *Dermacentor* (DAE100) than *Ixodes* (ISE6) cells, underpinning ecological vector differences ([PMID: 30140074](#)).

Section 15 — Model Organisms

- **Mammalian models:** Mouse (BALB/c, C57BL/6) and Fischer 344 rat are standard, challenged with attenuated LVS or virulent SCHU S4 (subsp. *tularensis*) ([PMID: 35077486](#), [PMID: 30054549](#), [PMID: 38713688](#)). The common marmoset has been used for inhalational challenge and therapeutic testing ([PMID: 41416830](#)).
- **Surrogate species:** *F. novicida* is a genetically tractable, lower-biosafety surrogate for mechanistic studies ([PMID: 35077486](#)).
- **Genetic models:** Bacterial gene-deletion/transposon mutants ($\Delta clpB$, $\Delta pilO$, $\Delta opiA$, *FTL_0883/FTT_0615c*) dissect virulence; host inflammasome knockouts (*Aim2*^{-/-}, *Nlrp3*^{-/-}) probe innate immunity ([PMID: 30054549](#), [PMID: 30057173](#), [PMID: 21670171](#), [PMID: 34690967](#), [PMID: 33875472](#)).
- **In vitro:** Human/murine macrophage, neutrophil, and dendritic-cell infection models; tick cell lines (DAE100, ISE6) for vector-competence studies ([PMID: 22357630](#), [PMID: 19841074](#), [PMID: 30140074](#)).
- **Phenotype recapitulation:** Rodent models reproduce intracellular replication, dissemination, granulomatous pathology, and lethality; SCHU S4 models severe type A disease, LVS models attenuated type B. **Limitation:** these models emphasize pneumonic/systemic lethality rather than the specific localized *glandular* lymphadenitis phenotype, which is best characterized from human case series.

Mechanistic Model / Interpretation



The unifying synthesis (Finding F013) is that **diagnostic delay driven by beta-lactam failure is the chief modifiable determinant of glandular tularemia morbidity**. The organism's intrinsic beta-lactam resistance, frequent culture-negativity, and low clinical familiarity conspire to delay correct diagnosis. Because intrinsic lethality of type B glandular disease is low, the dominant morbidity driver is **time-to-correct-diagnosis**, not disease severity per se. The actionable lever is clinical suspicion (endemic exposure + regional lymphadenopathy + beta-lactam non-response) confirmed by MAT ($\geq 1/160$) and/or PCR on node aspirate.

Evidence Base

PMID	Title (abbrev.)	Supports
38294108	<i>Systematic Review: Clinical Features, Treatment, Outcomes 1993–2023</i>	Glandular as major form; fatality by drug class (0.7/0.9/1.2%)
42641375	<i>Tularemia: a re-emerging zoonosis</i>	Low infectious dose, 30% untreated mortality, 2025 CDC first-line therapy, no vaccine, re-emergence data
42601640	<i>Tick-borne tularaemia in children</i>	Arthropod-borne → glandular/ulceroglandular; suppuration/surgery
16836829	<i>Epidemiologic/molecular analysis US 1964–2004</i>	Type A vs B differ in populations/site/geography
19251856	<i>Phylogeography of <i>F. tularensis</i></i>	Circumpolar type B; clonal radiation
23517149	<i>German isolates from brown hares</i>	European human infection from <i>Lepus europaeus</i>
30057173	<i>PI3-kinase effector OpiA</i>	Phagosomal escape mechanism
30054549	<i>ClpB mutants</i>	T6SS function/intracellular replication
35077486	<i>PilO virulence determinant</i>	Adherence/escape/virulence; mouse model
34690967	<i>Nlrp3 increases susceptibility</i>	Inflammasome host susceptibility
23763361	<i>Cytopathology of cervical lymphadenitis</i>	Granulomatous/suppurative histology; FNA role
10618057	<i>PCR vs culture</i>	PCR 75% vs culture 62% sensitivity
22090310	<i>Central Anatolia cases</i>	MAT $\geq 1/160$; beta-lactam failure
23104256	<i>Tularemia in central Anatolia</i>	Early treatment better; 30.9% failure; drainage 39–65%
22357630	<i>Neutrophil apoptosis inhibition</i>	Innate immune evasion
19841074	<i>CD14 and DC evasion</i>	CD14-dependent detection; DC cytokine evasion
24280916	<i>Tularemia — US 2001–2010</i>	1,208 cases; bimodal age/sex
1305858	<i>The ecology of tularaemia</i>	

PMID	Title (abbrev.)	Supports
		>250 species; two lifecycles; Northern Hemisphere
18755386	<i>Tularemia</i> (review)	Vectors/reservoirs; acute onset
33970545	<i>Innate immune kinetics / chloride transporter</i>	LVS limitations as vaccine/model
38713688	<i>Glucan-particle subunit vaccine</i>	Partial protection; F344 rat/SCHU S4 model
15677845	<i>Bichat guidelines</i>	Seven clinical forms; PEP agents; no person-to-person spread
41026652	<i>CDC 2025 treatment/PEP recommendations</i>	Updated treatment/PEP guidance
40907418	<i>Alsace 2024 increase</i>	9/10 not initially suspected; misdiagnosis
19801257	<i>Tularaemia mimicking glandular TB on adalimumab</i>	Immunosuppression as severity factor; histology
38294118	<i>Severe respiratory type B, fluoroquinolones</i>	FQ effective in severe disease
40466250	<i>German case series</i>	(Ulceroglandular most common; misdiagnosis; FQ/doxy resolution

Limitations and Knowledge Gaps

- 1. Form-specific data are limited.** Much evidence pools glandular with ulceroglandular disease; pure-glandular-only epidemiology, natural history, and QoL metrics are sparse.
- 2. No disease-specific QoL instruments** (EQ-5D/SF-36/PROMIS) have been applied to tularemia.
- 3. Mechanistic studies emphasize pneumonic/systemic lethality** (mouse/rat SCHU S4/LVS) rather than the localized glandular lymphadenitis phenotype; some mechanistic steps (e.g., precise inflammasome contribution to nodal granuloma formation) are **inferred** from systemic models.

4. **Culture-based data underrepresent true incidence** given ~10% culture positivity; surveillance likely undercounts.
5. **No human genetic susceptibility work** — whether host immunogenetic variation modulates glandular vs other forms is essentially unstudied.
6. **Vaccine gap:** No licensed human vaccine; LVS attenuation mechanism remains poorly defined.

Proposed Follow-up Actions

1. **Form-stratified cohort analysis:** Extract glandular-only cases from the 870-case systematic review and national registries to quantify pure-glandular incidence, suppuration rate, relapse, and time-to-diagnosis.
 2. **Diagnostic-delay intervention study:** Test whether endemic-area clinician decision aids (regional lymphadenopathy + beta-lactam non-response prompt) reduce time-to-appropriate-therapy and suppuration rates.
 3. **Doxycycline vs fluoroquinolone relapse comparison** in localized glandular disease, powered for relapse as the primary endpoint.
 4. **Node-localized mechanistic model:** Develop an intradermal/regional-node challenge model to study inflammasome (AIM2/NLRP3) and Th1 contributions specifically to granulomatous lymphadenitis.
 5. **Host immunogenetics:** Explore whether TLR4/CD14/inflammasome polymorphisms influence clinical form or severity.
 6. **Advance subunit vaccine candidates** (glucan-particle LPS+FTT0814; TMV OmpA/DnaK/Tul4) toward challenge studies against type A.
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Report compiled from 13 confirmed findings and 46 reviewed papers over 5 investigation iterations. Evidence classes: human clinical (case series, systematic reviews, surveillance), model organism (mouse/rat/marmoset), in vitro (macrophage/neutrophil/DC/tick-cell), and computational/phylogenomic.
